2016 Summer Qual

III. $\Sigma = LL^T$: cholesky decomposition, L : full column rank

$$y = Lz \sim N_n(0, \Sigma), \quad z \sim N(0, I)$$

$$y^T A y = z^T \underbrace{L^T A L}_{B} z$$

$$B^T = (L^T A L)^T = L^T A L = B \quad : \text{symmetric}$$

$$LL^T A LL^T A LL^T = LL^T A LL^T$$

Since L is full column rank, $\exists (L^T L)^{-1}$

$$\cancel{(L^T L)^{-1} L^T L} L^T A L L^T A L L^T = \cancel{(L^T L)^{-1} L^T L} L^T A L L^T$$

$$\Rightarrow L^T A L L^T A L L^T = L^T A L L^T$$

$$\Rightarrow L^T A L L^T A L \cancel{L^T L (L^T L)^{-1}} = L^T A L L^T \cancel{L (L^T L)^{-1}}$$

$$\Rightarrow L^T A L L^T A L = L^T A L$$

$$\Rightarrow L^T A L \text{ : idempotent}$$

$$\text{rank}(L^T A L) = tr(L^T A L) = tr(A L L^T) = tr(A\Sigma) = r$$

$$z^T B z - y^T A y \sim \chi^2_r$$

$\blacksquare$

3.

$$\hat{\beta}^* = (X_1^T X_1)^{-1} X_1^T Y \Rightarrow E[\hat{\beta}^*] = (X_1^T X_1)^{-1} X_1^T [X_1 \beta_1 + X_2 \beta_2]$$

$$= \beta_1 + (X_1^T X_1)^{-1} X_1^T X_2 \beta_2$$

bias of $\hat{\beta}^* = (X_1^T X_1)^{-1} X_1^T X_2 \beta_2$

$$\mathrm{var}(x_{01}^T \hat{\beta}^*) = \sigma^2 x_{01}^T (X_1^T X_1)^{-1} x_{01}$$

$$\mathrm{var}(x_0^T \hat{\beta}) = \sigma^2 x_0^T (X^T X)^{-1} x_0$$

$$= (x_{01}^T \; x_{02}^T) \begin{pmatrix} X_1^T X_1 & X_1^T X_2 \\ X_2^T X_1 & X_2^T X_2 \end{pmatrix}^{-1} \begin{pmatrix} x_{01} \\ x_{02} \end{pmatrix}$$

$$= (x_{01}^T \; x_{02}^T) \begin{pmatrix} G^{11} & G^{12} \\ G^{21} & G^{22} \end{pmatrix} \begin{pmatrix} x_{01} \\ x_{02} \end{pmatrix}$$

$$G^{11} = (X_1^T X_1)^{-1} + (X_1^T X_1)^{-1} X_1^T X_2 (X_2^T (I - H_{X_1}) X_2)^{-1} X_2^T X_1 (X_1^T X_1)^{-1}$$

$$G^{12} = -(X_1^T X_1)^{-1} X_1^T X_2 [X_2^T (I - H_{X_1}) X_2]^{-1}$$

$$G^{21} = -[X_2^T (I - H_{X_1}) X_2]^{-1} X_2^T X_1 (X_1^T X_1)^{-1}$$

$$G^{22} = [X_2^T (I - H_{X_1}) X_2]^{-1}$$

$$\mathrm{var}(x_0^T \hat{\beta}) - \mathrm{var}(x_{01}^T \hat{\beta}^*)$$

$$= (x_{02} - X_2^T X_1 (X_1^T X_1)^{-1} x_{01})^T \underbrace{G^{22}}_{\text{psd}} (x_{02} - X_2^T X_1 (X_1^T X_1)^{-1} x_{01})$$

$$\geq 0$$

$\boxtimes$

4. For $cov(e) = \frac{\sigma^2}{2}(I+J)$,

$$Y^* = \sqrt{2}(I+J)^{-1/2} Y$$

$$X_c^* = \sqrt{2}(I+J)^{-1/2} X_c$$

$$1^* = \sqrt{2}(I+J)^{-1/2} 1$$

$$\frac{\partial \|Y^* - 1^*\alpha - X_c^*\beta_c\|^2}{\partial \alpha} = 0$$

$$\Rightarrow -2 1^{*T}(Y^* - 1^*\alpha - X_c^*\beta_c) = 0$$

$$1^{*T} Y^* = 1^{*T} 1^* \alpha$$

Using Shermann - Morrison Lemma

$$1^{*T} 1^* = 2 1^T (I+J)^{-1} 1 \quad \& \quad 1^{*T} Y = 2 1^T (I+J)^{-1} Y$$

$$= 2 1^T (I + 11^T)^{-1} 1 \qquad = \frac{2}{n+1} \Sigma y_i$$

$$= 2 \cdot 1^T \left(I - \frac{1}{1+n} J\right) 1$$

$$= \frac{2n}{n+1}$$

$$\hat{\alpha} = \frac{2}{n+1} \Sigma y_i \cdot \frac{n+1}{2n} = \bar{y}$$

$$\frac{\partial \|Y^* - 1^*\alpha - X_c^*\beta_c\|^2}{\partial \beta_c} = 0$$

$$\Rightarrow -2X_c^{*T}(Y^* - 1^*\alpha - X_c^*\beta_c) = 0$$

$$\hat{\beta}_c = (X_c^{*T}X_c^*)^{-1}X_c^{*T}Y^*$$

$$= \frac{n+1}{2n} \cdot \frac{2n}{n+1}(X_c^TX_c)^{-1}X_c^TY$$

$$= (X_c^TX_c)^{-1}X_c^TY$$

$\therefore$ Same. $\blacksquare$